
NLP

ASSIGNMENT 3

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Text Classification Using RNN, LSTM, Transformer, and Pretrained Language Model

- **Preprocessing:** After loading the csv file and dropping any rows with missing values, the following preprocessing steps are used:
 - Converting it to lowercase
 - Removing the URLs
 - Change multiple dots to single dot
 - Remove extra white spaces
 - Remove non-alphabetical characters.
 - Removed the stop words using https://github.com/TrigonaMinima/HinglishNLP/blob/master/data/assets/stop_hinglish
- **RNN and LSTM:**

Both models are implemented using PyTorch. It produces class predictions by first creating an embedding layer, then a single RNN/LSTM layer, followed by a dropout layer.
- **Transformers:**

Transformer models use token embeddings and positional encodings to form the encoder layer, average the encoder outputs, apply dropout, and then make the final prediction.
- **BERT:**

Uses the pretrained BERT model: "l3cube-pune/hing-bert" (<https://huggingface.co/l3cube-pune/hing-bert>), which is a BERT pre-trained model on Hindi-English code-mixed on Roman text. *The BERT model was not included in the GitHub due to its large size*

All the models use Adam optimizers with a learning rate of 0.001 for RNN, LSTM, and transformers. For BERT, the learning rate is 0.00002. All models are trained for 30 epochs with an early stopping criterion for patience=10 and delta = 0.001 to prevent overfitting.

Model	Accuracy	macro precision	macro recall	macro F1-score
RNN	0.6	0.52	0.52	0.52
LSTM	0.61	0.52	0.52	0.51
Transformer	0.62	0.53	0.53	0.53
BERT	0.67	0.64	0.65	0.64

Table 1: Performance metric for hate dataset

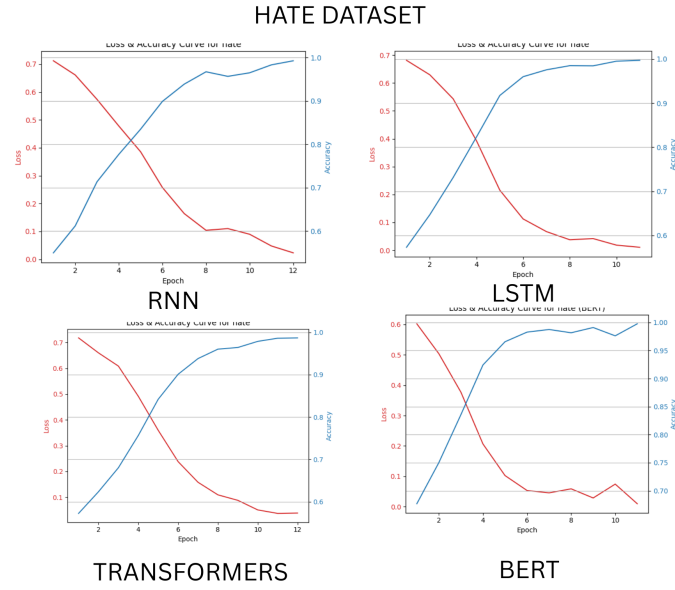


Figure 1: Training accuracy and loss plot for hate dataset

Model	Accuracy	macro precision	macro recall	macro F1-score
RNN	0.91	0.77	0.63	0.67
LSTM	0.91	0.76	0.62	0.65
Transformer	0.94	0.82	0.87	0.84
BERT	0.97	0.88	0.94	0.91

Table 2: Performance metric for sarcasm dataset

Model	Accuracy	macro precision	macro recall	macro F1-score
RNN	0.52	0.52	0.52	0.51
LSTM	0.54	0.61	0.59	0.53
Transformer	0.56	0.54	0.54	0.54
BERT	0.66	0.65	0.65	0.65

Table 3: Performance metric for humor dataset

SARCASM DATASET

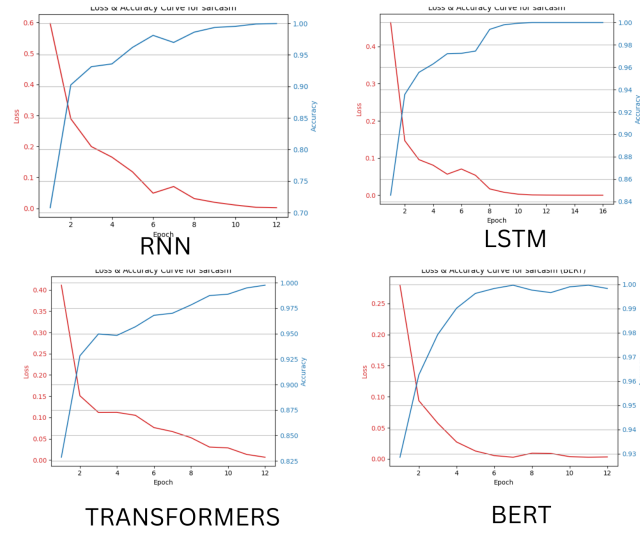


Figure 2: Training accuracy and loss plot for sarcasm dataset

HUMOR DATASET

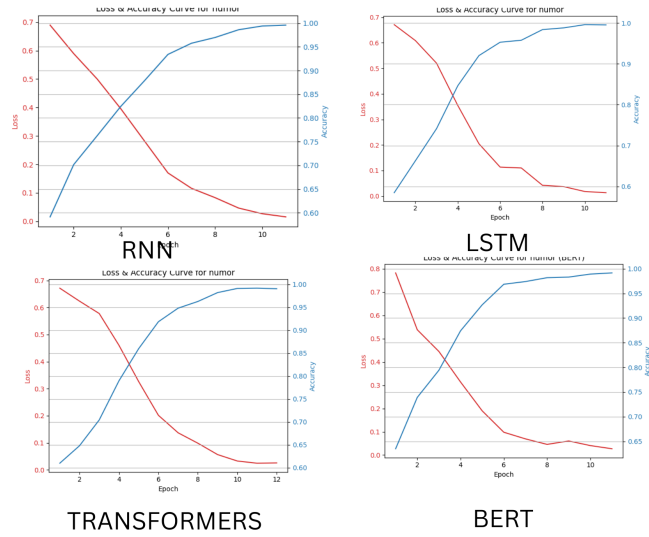


Figure 3: Training accuracy and loss plot for humor dataset